Porosity *(f)* = Volume of voids (*V*f) / Total volume of soil (*V*t)
= (*V*a + *V*w) / (*V*s + *V*a + *V*w)

Where: *V*s = Volume of solids, *V*a = Volume of air, *V*w = Volume of water

Porosity values generally range from 0.3 to 0.6 (30–60 %). Coarse-textured soils tend to be less porous than fine-textured soils. If both bulk density and particle density are known, the total porosity can be calculated using the formula:

$$\text{Total porosity} = \frac{\text{Particle density (ps)} - \text{Bulk density (pb)}}{\text{Particle density (ps)}} \times 100\ \%$$

4.5.1.8 Void ratio

The void ratio is also an index of the fractional pore space in relation to the volume of solids. Void ratio is given by:

Void ratio (*e*) = Volume of voids (*V*f) / Volume of solids (*V*s)
= (*V*a + *V*w) / (*V*t − *V*f)

Where *V*a = Volume of air, *V*w = Volume of water, *V*t = Total volume of soil, *V*s = Volume of solids, *V*f =Volume of voids

4.5.1.9 Soil water content

The water content of a soil can be expressed on mass or volume basis.

4.5.1.9.1 Gravimetric moisture content (mass wetness)

Gravimetric moisture content refers to the mass of water relative to the mass of dry soil particles.

$$\text{Gravimetric moisture content } (\theta g) = \text{Mass of water } (Mw)/\text{Mass of solids } (Ms) = \frac{\text{Mass of wet soil} - \text{Mass of oven dry soil}}{\text{Mass of oven dry soil}}$$

Gravimetric moisture content is sometimes expressed as a decimal fraction but more often as a percentage.

4.5.1.9.2 Volumetric moisture content (volume wetness)

Volumetric moisture content (Θv) = Volume of water (*V*w)/ Total volume of soil (*V*t) Thus;

$$\Theta v = \text{Volume of water } (Vw) / [\text{Volume of solids } (Vs) + \text{Volume of voids } (Vf)]$$

The volumetric water content refers to the volume of water relative to the total soil volume. At saturation, the volumetric moisture content is equal to the porosity.

Note that volumetric moisture content = gravimetric moisture content x bulk density

4.5.1.10 Degree of saturation

The degree of saturation expresses the water volume present in the soil relative to the pore volume. It is given by;

Degree of saturation (s) = Volume of water (Vw) / Volume of voids (Vf)
= Volume of water (Vw) / [Volume of air (Va) + Volume of water (Vw)]

The degree of saturation ranges from zero in a completely dry soil to 100 % in a saturated soil.

4.5.1.11 Air-filled porosity

Air-filled porosity is a measure of the relative content of air in the soil. It is the volume of air-filled pore space in a soil at given moisture content. This parameter is an important criterion of soil aeration. Air-filled porosity is negatively related to the degree of saturation thus;

Air-filled porosity (fa) = porosity (f) – degree of saturation (s)
Air-filled porosity is also given by;
$f\text{a} = V\text{a}/V\text{t} = V\text{a}/(V\text{s} + V\text{a} + V\text{w})$

Hillel (2004) in the book Introduction to Environmental Soil Physics provides more information on soil water content.

2.5.9 The soil profile

In often cases, soil scientists dig a large hole (pit) which is usually several meters deep and about a meter wide to expose soil layers for study. The vertical section which exposes a set of soil layers in the wall of the pit is termed as the soil profile. A soil profile consists of a succession of horizontal layers of different properties known as soil horizons. A soil profile tells you much about the environment and history of a region and warns you about the potential problems of using the land.

The horizons are approximately parallel to the land surface and several layers may evolve simultaneously over a period of time. Horizons within the soil profile somewhat have irregular boundaries and may vary in thickness. The soil horizons differ from one in their physical, chemical, and biological properties. They provide information on colour, texture, structure, permeability, drainage and biological activity. Six main groups of soil horizons have been identified and are designated symbols O, A, E, B, C and R. The three major layers that make up a soil profile are A, B and C horizons. Sometimes, the O and E horizons are absent.

O horizons lie on the surface, above the mineral horizon. They contain loose and partly decayed organic matter or organic deposit with litter layer of plant residues in relatively non-decomposed form. It represents the shallow or thick surface zone of decaying organic debris.

A horizons are the topmost mineral horizons lying below the O horizon. It refers to the surface soil which contains large amounts of mineral matter (sand, silt and clay) mixed with humus. It

is a layer of mineral soil with the most organic matter accumulation and microbial activities. This horizon is of biological concern as it contains most of the roots, small animals and microorganisms. The high organic matter content of this horizon makes it a dominant reservoir of microbial food. The A horizon is darker in colour than the horizons below because the humus has a black or dark-brown colour. In the absence of the O horizon, you may find the A horizon as a surface horizon.

E horizons are mineral horizons located under the A horizon. They are zones of maximum leaching and eluviations (washing out) and consist of light coloured mineral particles. The E horizons are significantly leached of clay, humus, iron, and aluminium oxides, which leaves a concentration of resistant minerals, such as quartz, in the sand and silt sizes. These horizons are commonly grayish in colour and contain low organic matter.

B horizons are located underneath E horizons which in the absence of the E horizon lie directly below the A horizon. B horizons are zones of illuviation (washing in / accumulation) of soil materials removed from A and E horizons. These horizons reflect chemical or physical alteration of parent material and accumulate iron, clay, aluminium and organic compounds transported from E horizons. In addition, B horizons contain little organic matter and few plant roots.

C horizons are located under the B horizons and contain the parent material of the soil. The C horizons are found at the bottom of the soil profile. They contain mixtures of weathered rocks and minerals which are less affected by soil-forming processes and lacks pedological (soil) development.

R horizons are underlying solid rocks with little evidence of weathering.

The altered layer of soil above the parent material that includes the surface and subsoil layers is called solum.

Figure 4 shows you the horizons in the soil profile;

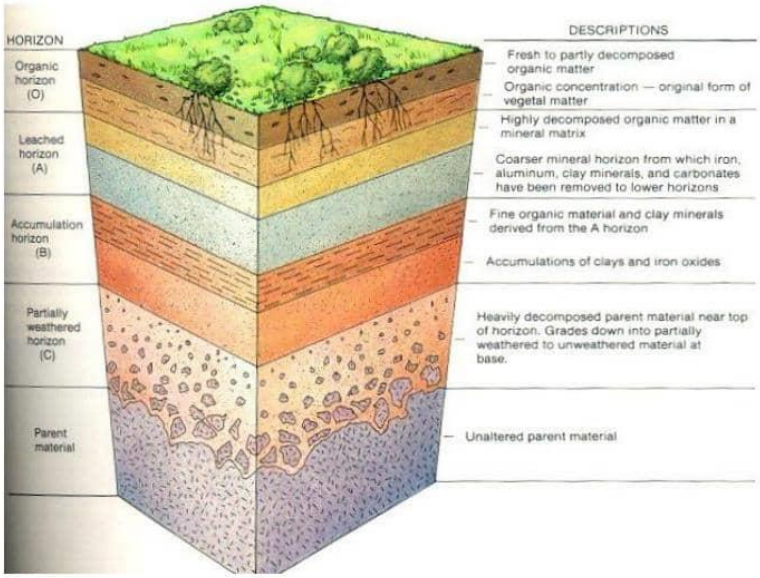


Figure 4: The soil profile
Source: https://i.pinimg.com/originals/e4/3c/07/e43e070ff246646febdb3938bbd8a8f4.jpg

Transition Horizons are horizons where properties of two master horizons merge. Transition horizons are designated by combining the two master horizon symbols, for example, AB, BA, BC and CB.

Mixed Horizons are intermingled parts of two master horizons to the extent that they cannot be separated. Mixed horizons are designated by combining the two master horizon symbols. For example, A/B, B/C or C/B.

4.5 Unit Topics

4.5.1 Physical properties

Soil physical properties affect the appearance and feel of a soil. They include texture, structure, consistency, density, porosity and water content. These properties are dominant factors which affect the use of soils. Soil profile and the horizon features are often described in the field using the physical properties.

4.5.1.1 Texture

4.5.1.1.1 Defining soil texture

What is soil texture? It is the relative proportion of particles of various sizes such as sand, silt According to (Jones, 2012), the U.S system of classification for the various proportions is;

Sand = 2 to 0.05 mm, Silt = 0.05 to 0.002 mm and Clay = <0.002 mm

Why is it important to know the texture of soils?

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It is important to know the soil texture because it controls water movement in soil, influences chemical reactivity and nutrient availability, and is a factor in the erosion potential of a soil. Coarser textures will have larger packing voids and, therefore, larger pore spaces; water can move and drain very rapidly through these soils. The greater surface area and surface charge provided by finer textures results in the soil holding more water and having a higher ability to store more nutrients. Finer textures usually have greater resistance to erosion.

In addition, texture is a major tool in determining crop suitability and approximating soil response to environmental and management conditions such as drought, liming and workability. For example, clay soils will be sticky when wet and hard when dry.

Note that for all mineral soils, the proportions of sand, silt, and clay always adds up to 100 percent. These percentages are grouped into soil texture "classes", which have been organized into a "textural triangle".

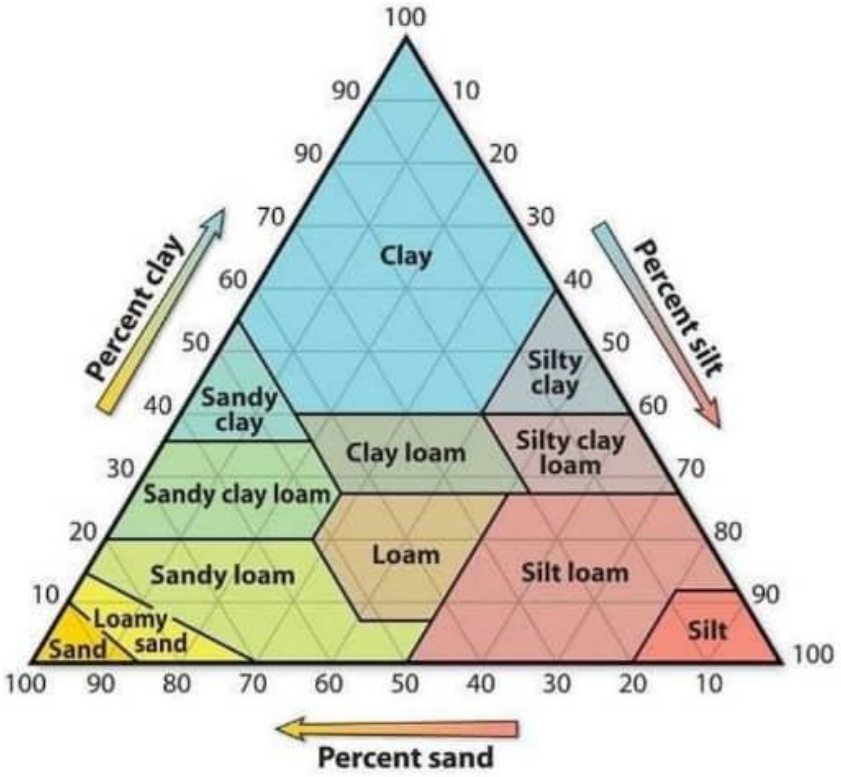


Figure 7: The textural triangle

Source: https://i.pinimg.com/originals/cd/39/4e/cd394e037f7375299257ff166ae594d1.jpg

Collectively, the soil separates of sand, silt, and clay are called the "**fine-earth fraction**", and represent inorganic soil particles less than 2 mm in diameter.

4.5.1.1.2 Particle size analysis

Particle size analysis also known as mechanical analysis is the determination of the size distribution of mineral particles in soils. It is the determination of relative proportions of sand, silt and clay. The mineral matter less than 2 mm in diameter is considered.

4.5.1.1.3 Methods of determining particle size distribution

4.5.1.1.3.1 Feel method

In this method, the soil is moistened with water and rubbed between the thumb and fingers. The way the wet soil "sticks out" of your thumb and fingers gives you a good idea of the clay

content. The sand particles are gritty, the silt has a floury/ smooth feel and is moderately plastic and sticky when wet. You can also push the soil between your thumb and forefinger to make a ribbon. The longer the ribbon, the more clay there is in the soil. This method is less accurate but much faster. Accuracy of the feel method largely depends on experience.

4.5.1.1.3.2 Separation by sedimentation

This type of analysis depends fundamentally upon **Stokes' Law**. This law states that denser (larger) particles sink farther than less dense (smaller) particles when suspended in a liquid.



Reflection

Make a soil suspension (a mixture of soils and water) in a container. Agitate (shake) the soil suspension and observe the rate at which particles of various sizes in the mixture settle. Which particles settle out faster between larger and smaller constituents?

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Stokes Law

The rate of fall (v) of a particle in a suspension can be predicted from Stokes' Law:

$v = d^2 (\rho s - \rho l)\, g/18n$

Where:

v = velocity of particle falling in a liquid, cm/sec

d^2 = diameter of particles, cm

ρs = density of the solid particles, ~2.65 g/cm^3

$\rho 1$ = density of the liquid, g/cm^3

g = acceleration due to gravity, 980 cm/sec^2

n = viscosity of the liquid = 1 $g.cm^{-1}.s^{-1}$

The formula shows that the settling velocity, v, is directly proportional to the square of the particle's effective diameter, d; the acceleration of gravity, g; and the difference between the

density of the particle, ρs, and density of the liquid, ρl; but inversely proportional to the viscosity (resistance to flow) of the liquid, η.

Note that calculations using this law are only approximately correct for soils because particles are not spherical and vary in density.

The density of water and its viscosity both change in such a manner that particles settle faster with increased temperature.

Dispersion

As you know, soils naturally exist as aggregates and not as a dispersed mixture of particles. The soil aggregates are formed by cementing a mixture of primary particles; sand, silt, and clay by cementing agents such as organic matter, mineral oxides, or polyvalent cations (Ca^{+2} and Al^{+3}). Dispersive methods remove or inactivate these binding agents in order to perform accurate analysis of the settling rate of individual particles.

Dispersion can be achieved using mechanical or chemical means. Mechanical stirring overcomes weaker binding forces in large aggregates. Chemical agents deflocculate (separate) clays by replacing polyvalent cations which bind the clays by monovalent cations thereby breaking the inter-particle linkage. e.g sodium metaphosphate.

Three (3) methods which include (i) pipette method, (ii) successive sedimentation method, and (iii) hydrometer method have been discussed.

4.5.1.1.3.2.1 Pipette method

The pipette method is often used as the standard to which other methods are compared. This method involves removing the carbonates by adding hydrochloric acid. Organic matter is removed by destroying it with an oxidizing agent such as hydrogen peroxide. Iron oxides, chlorides and sulphates are also removed. The soil sample is then dispersed using sodium metaphosphate. Sands are separated by wet sieving. Clay is determined by sampling a suspension with a pipette at a specific depth and time following stokes law. Silt content is found as the difference between total clay and silt and clay.

Silt = (Silt + Clay) - Total Clay

4.5.1.1.3.2.2 Successive sedimentation method

Soil particles are dispersed, sand and silt are allowed to settle followed by removal of clay by decanting. Sand is separated from silt by wet sieving and clay content is determined by subtracting weight of sand and silt from total weight of sample.

4.5.1.1.3.2.3 Hydrometer method

The hydrometer method is somewhat less accurate than the pipette method, but is easier and faster to perform. The hydrometer method of mechanical analysis relies on the principles of dispersion and sedimentation. The soil particles are dispersed using sodium metaphosphate. The amount of each type of particle is determined using Stoke's law, the principle that larger, heavier particles (such as sand) will settle out of water faster than smaller particles. This method is based on the change of density of a soil and water suspension upon the settling of the soil particles. The instrument used for measuring the density of the suspension is called hydrometer.

The hydrometer is calibrated to measure the density of a suspension in units of grams per liter. Research has determined that sand particles settle within 40 seconds. Therefore, at 40 s, the density of clay and silt is determined. The density at 8 hours is used to determine percent clay.

% (Silt + Clay) = Density at 40s x 100/ mass of soil
% Sand = 100 % - % (Silt + Clay)
% Clay = Density at 8 hours x 100/ mass of soil
% Silt = 100 % - % Sand - % Clay
Note that the volume of the suspension is taken to be 1 liter. Therefore,
Mass = density (g/l) x volume (l), volume = 1 liter, therefore, mass = density

4.5.1.1.4 Major textural grouping of soils

After analysis of particle size distribution, soils are grouped into textural classes. The textural triangle is used to assign textural class names. The textural triangle contains twelve textural classes. These are;

- Clay, sandy clay, silty clay
- Sandy clay loam, clay loam, silty clay loam, loam, sandy loam, silty loam
- Loamy sand, sand
- Silt

Note that soils within each textural class have similar physical and chemical properties. The textural class name reflects the relative influence of each soil separate (sand, silt and clay) on the properties of soil.

Clayey textured soils

Because of its small size and sheet-like structure, clay has a large amount of surface area per unit mass, and its surface charge attracts ions and water. Therefore, clayey soils have higher nutrient and water holding capacity. They feel slippery and are usually too sticky when wet and too hard when dry to cultivate. These soils may have shrink-and-swell characteristics that affect construction uses.

Sandy textured soils

These separates are commonly comprised of quartz or some other inactive mineral. Sandy soils feel gritty and are low in fertility due to nutrient leaching. These soils have low buffering and cation exchange capacity. They are rapidly permeable and hold less soil moisture.

Loamy textured soils

Loamy soils have properties resulting from about equal influences of sand, silt and clay. They permit slower movement of water and are better able to retain moisture and nutrients. Loamy soils feel crumbly and are generally more fertile. They have higher cation exchange and buffer capacities.

Silt

Silt particles have intermediate size and chemical and physical properties between clay and sand. They have limited ability to retain nutrients. They tend to have spherical shape which makes the soil to feel slippery or soapy. These soils are easy to cultivate as particles do not have the ability to stick together.

Texture is considered to be a permanent characteristic of a soil because cultivation and other management practices do not alter the sizes of individual soil particles. However, erosion and deposition may rapidly alter the particle size distribution.

4.5.1.2 Soil structure and aggregation

4.5.1.2.1 Defining soil structure

The sand, silt and clay particles of the soil are cemented together in the soil to form natural units of compound particles or aggregates separated by pores, cracks or planes of weakness. Soil structure is the arrangement and organisation of particles (sand, silt, and clay) and other soil materials into distinct aggregates. The structure of soil is not a permanent or stable property but is subject to change under the influence of factors such as climate, biological activity and tillage.

Why is it important to know the structure of the soil? It is important to know the structure of soils because it determines the total porosity of the soil and affects the shapes and sizes of pores in the soil thereby affecting infiltration and aeration. Soils with good structure have relatively large pore spaces that occur between peds and this facilitates water and air movement. Soil structure also affects properties that affect germination, root growth, tillage erosion as well as the ability of the soil to support roads and traffic.

4.5.1.2.2 Classification of soil structure

In general, there are two broad classes.

1. Non aggregated structure
2. Aggregated structure

4.5.1.2.2.1 Non-aggregated structure

In this group, two types of structures are distinguished.

i. **Single grained structure**

Refers to soil composed of loose grains of primary particles that are not cemented together. Sand grains are an example of such a structural type.

ii. **Massive structure**

This refers to soil in which particles are tightly packed into large cohesive mass or block that does not have natural lines of weakness. This type of structure is usually common in soils containing high clay content. The soils occur as large clods without clearly defined characteristic shape.

4.5.1.2.2.2 Aggregated structure

Aggregated structures refer to conditions in which soil particles exist as secondary particle formed by cementing of several primary particles. The secondary particles are called aggregates/peds. The peds are separated from each other by surfaces of natural weakness, which define their boundaries. When soils are disturbed, they tend to break along these natural boundaries.

When describing aggregated structure, 3 properties are used;

i. The shape of aggregate (ped) called the **type.**
ii. The size of the aggregate/ped called **class.**
iii. The distinctness of the aggregates or how easily the aggregates can be identified or seen in the field. This property is called **grade**.

Types of structure for aggregated soils

Aggregated structures are grouped into 3 major groups. Broadly described as (i) plate-like (ii) prism-like and (iii) block-like

i. Plate-like → Platy structure

ii. Prism-like → Columnar / Prismatic

iii. Block-like → Spherical → Crumb
Spherical → Granular
Block-like → Flat surfaces → Angular blocky
Flat surfaces → Sub-angular blocky

Type	Description
Single grain	No aggregation of coarse particles when dry
Granular	Rounded surfaces (porous)
Crumb	Rounded surfaces but larger than granular (very porous)
Blocky	Cube-like with flattened surfaces and sharp corners
Sub-angular blocky	Cube-like with flattened surfaces and rounded corners
Prismatic	Rectangular with a long vertical dimension and flattened top
Columnar	Rectangular with a long vertical dimension and rounded top
Platy	Rectangular with a long horizontal dimension

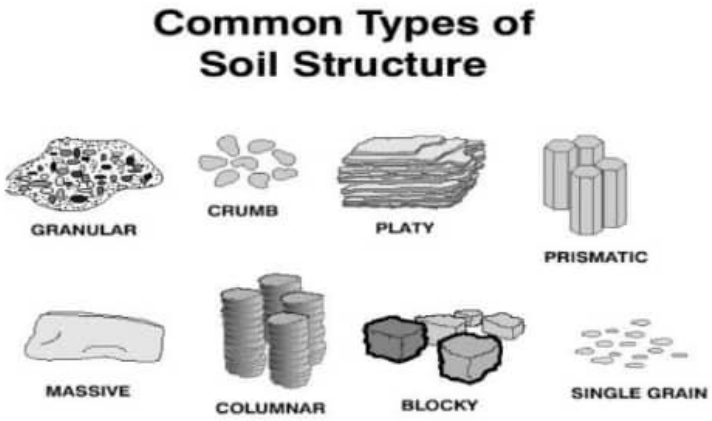


Figure 8: Types of soil structure

Source:http://41.77.4.164/image.slidesharecdn.com/ed-field-afg-purdue-unite-lesson4-texture-structure-ppt-140528173554-phpapp02/95/soil-texture-and-structure-17-638.jpg?cb=1401298593

Classes of soil structure

A number of descriptive terms are used to describe the size or class of soil aggregates. These terms refer to specific size ranges which are however not uniform but vary with type of soil structure. The terms commonly used to describe the class of soil aggregates are;

Very fine/very thin
Fine/thin
Medium
Coarse/large
Very coarse/very large

↓

Increasing size of block

Grade of structure

The grade refers to the degree of distinctiveness of the peds. It also refers the degree of aggregation and expresses the differential between cohesion within aggregates and adhesion between aggregates. The grade depends on the moisture status of the soil. Aggregates are more distinct in dry soil than in moist soils. The terms used to describe the grade of the soil include;

Structureless	No observable aggregation or no definite orderly arrangement of natural lines of weakness. Massive if coherent; single grain if non-coherent
Weak	The degree of aggregation characterised by poorly formed indistinct aggregates that are barely observable in place.
Moderate	Well-formed distinct aggregates that are moderately durable and evident in a soil sample but not distinct in undisturbed soil.
Strong	Durable aggregates that are quite evident in undisturbed soil and adheres weakly to one another.

Compound structure

Sometimes you may find soil horizons that consist of compound structures. These structures comprise of one or more sets of smaller peds that are held together to form larger peds. For example, very coarse prismatic structure consisting of medium sub-angular blocky structure.

4.5.1.2.3 Aggregation and aggregate stability

4.5.1.2.3.1 Aggregation percentage

In the previous section, we defined soil structure as the arrangement and organisation of particles (sand, silt, and clay) and other soil materials into distinct aggregates. It is important for you to note that the aggregation percentage is closely related to the grade of soil structure

Aggregation percentage is determined by sieving an uncrushed air dry soil of known mass using a series of sieves with different diameters. The common series of sieves include a combination of diameters 8 mm, 5 mm, 3 mm and 2 mm and a collecting pan. Firstly, the whole soil sample is passed through 8 mm sieve followed by further sieving using sieves with smaller sizes. The mass of soil retained on each of the sieves and the mass of particles retained on the collecting pan are weighed. Following this is the expression of the mass retained on each sieve and the receiver as a percentage of total mass of dry soil that passed through the 8 mm sieve. Finally, the quality of aggregation is expressed by mean weight diameter (MWD) which is defined by the formula below;

$$MWD = \sum X_i W_i$$

Where

X_i is the mean diameter of soil particles considering the two limits of sieves between which sizes of aggregates falls.

W_i is the weight of aggregates retained between two sieve diameters expressed as a fraction of total mass of soil analysed.

4.5.1.2.3.2 Aggregate Instability Index (AII)

Perhaps you are wondering, what is Aggregate Instability Index? The aggregate instability index is the measure of how stable or unstable soils aggregates are when subjected to mechanical agitation under wet conditions. To determine AII, soil aggregates are sieved under dry and wet conditions. The MWDs of both wet and dry are calculated and the difference between them is used to calculate the instability index.

$$AII = MWD_{dry\ soil} - MWD_{wet\ soil}$$

The greater the difference, the higher the AII, the more unstable is the soil structure. The lower the AII, the more stable the structure of the soil.

The following limits are used to interprete AII.
If AII is <1 structure is stable, 1-2 moderately stable, >2 structure is unstable

4.5.1.2.4 How to maintain soil structure

The causes of soil structure deterioration include compaction, cultivation, de-vegetation, high sodium contents and excessive moving and handling of soil. Contrary, practices that prevent compaction, provide soil cover and enhance accumulation of organic matter result in improved soil structure.

Soil structure can be maintained through the following management strategies.

1. Tilling the soil only at the proper moisture contents. Do not till the land when the soils are too wet to avoid destroying the soil aggregates.
2. Applying lime to acid soils results in good plant growth and development of good soil structure.
3. Application of organic matter and growing grasses and legumes results in improved soil structure.

4.5.1.3 Soil colour

The colour of the soil is one physical property used to distinguish different soils in the field. It is used in a number of classification systems as one of the criteria for differentiating soils. Soil colour is useful because other more important characteristics that are not so easily quantified may be inferred (concluded) from it.

Some important information that you can infer from colour include,

1. Dark colour "values" indicate more organic matter than light colour "values".
2. Light gray or grayish colours commonly indicate reducing (low oxygen) conditions, either currently or in the past. This is common in soils that are usually saturated with water.
3. In some environments, yellowish or reddish mottles (spots) indicate alternating oxidizing (oxygen rich) and reducing (low oxygen) conditions. This is common in some wetland soils that are only seasonally or periodically saturated with water.
4. In well aerated soils, oxidized or ferric (Fe^{+3}) iron compounds are responsible for the brown, yellow, and red colours you see in the soil.

Soil colour is also helpful in distinguishing boundaries in differing soils (horizons) within a soil profile.

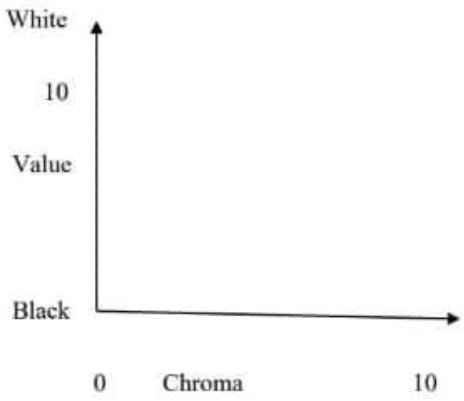
In order to standardize the method of describing colour of soils, a number of soil colour systems have been developed. The most widely colour reference chart used is Munsell soil colour chart. Using the Munsell system, colour is described in terms of 3 properties.

1. Hue
2. Value
3. Chroma

Hue; describes where in the colour spectrum the soil colour exists e.g. yellow, red, blue, green and gray.

Value; describes the lightness or darkness of the colour. The lower the numerical value, the darker the colour and the higher the value, the lighter it is.

Chroma; indicates the strength or brightness or distinctiveness of the colour.



In a Munsell notation, the colour is written in the order hue-value-chroma. The color "5YR 4/3" is an example of a Munsell notation, where 5YR is the hue, 4 is the value, and 3 is the chroma.

There are five (5) principal hues. These are Red (R), Yellow (Y), Green (G), Blue (B) and Purple (P). There complement/intermediate colours include Yellow-Red (YR), Green-Yellow (GY), Blue-Green (BG), Purple-Blue (PB) and Red-Purple (RP).

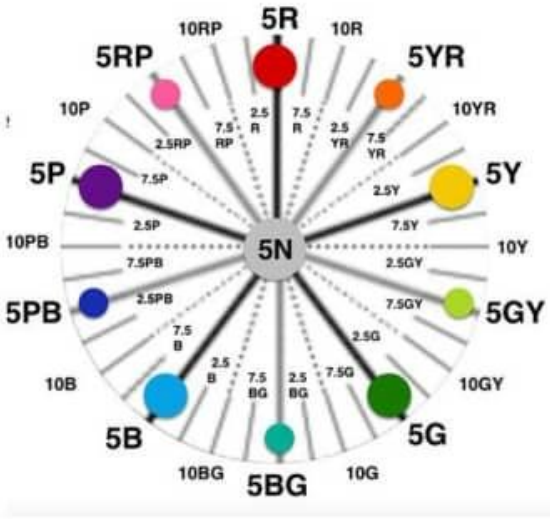


Figure 9: Standard Munsell hue circle divisions

Source:https://munsell.com/wp-content/uploads/2017/03/standard-munsell-hue-circle-divisions.jpg

Moisture content affects the colour of the soil therefore the need to indicate whether dry or moist. e.g 5YR 4/3" (moist). In most cases, wet soils appear darker than wet soils.

4.5.1.4 Soil consistency

4.5.1.4.1 What is soil consistency?

Consistency refers to the soils behavior when subjected to stress. It describes the resistance of a soil at various moisture contents to mechanical stress or manipulation. Soil consistence gives you an idea about forces of cohesion and adhesion that are holding the aggregates together. It indicates the amount and type of clay material, tillage conditions and the potential for compaction. Further, soil consistency helps to predict cultivation problems and gives you an idea about the soils ability to support applied forces such as building weight, vibration or road traffic. Consistency is measured at three (3) moisture conditions; air dry, moist and wet.

4.5.1.4.2 Consistency when dry

Under dry soil conditions, consistency is measured by the soils resistance to rapturing or to fragmentation when squeezed. The following terms are used to describe consistency under dry soil conditions: loose, soft, slightly hard, hard, very hard, and extremely hard.

Loose	Non-coherent
Soft	Soil mass is very weakly coherent and breaks to individual grains under very slight pressure.
Slightly hard	Weakly resistant to pressure; easily broken between thumb and forefinger.
Hard	Moderately resistant to pressure; can be broken in the hands without difficulty but is barely breakable between thumb and forefinger.
Very hard	Very resistant to pressure; can be broken in the hands only with difficulty; no breakable between thumb and forefinger.
Extremely hard	Extremely resistant to pressure; cannot be broken in the hands

4.5.1.4.3 Consistency when moist

Under moist soil conditions, the resistance to sheering or squeezing forces by the thumb or forefinger is observed. The following terms are used to describe consistency under moist conditions: very friable, friable, firm and very firm.

Very friable	Soil material crushes under very gentle pressure but coheres when pressed together
Friable	Soil material crushes easily under gentle to moderate pressure
Firm	Soil material crushes under moderate pressure
Very firm	Soil material crushes under strong pressure; barely crushable between thumb and forefinger
Extremely firm	Soil material only crushes under very strong pressure, cannot be crushed between thumb and fore-finger.

4.5.1.4.4 Consistency when wet

In wet conditions, the soils plasticity (ability to be molded) and its stickiness are measured. Plasticity of a soil is the ability of soil material to change shape continuously under the influence of an applied stress and to retain the impressed shape on removal of the stress. Stickiness is the quality of adhesion of the soil material to other objects.

The following terms are used to describe consistency under wet soil conditions:

4.5.1.4.4.1 Stickiness

The following terms are used to describe stickiness; non-sticky, slightly sticky, sticky, very sticky.

Non-sticky	After release of pressure, no soil material adheres to thumb or finger.
Slightly sticky	Soil material adheres to both thumb and finger but comes off one or the other rather cleanly.
Sticky	Soil material adheres to both thumb and finger and tends to stretch somewhat and pull apart.
Very sticky	Soil material adheres strongly to both thumb and finger and stretches after separation.

4.5.1.4.4.2 Plasticity

Plasticity is described by terms such as non-plastic, slightly plastic, plastic and very plastic.

Non-plastic	No wire is formable
Slightly plastic	wire formable but soil mass easily deformable
Plastic	wire formable and pressure required for deformation of the soil mass

Very plastic	wire formable and much pressure required for deformation of the soil mass

4.5.1.5 Particle density

Particle density measures the density of soil particles only. The measurement of particle density does not include the mass of water or pore space. The formula for calculating particle density is:

Particle density (ρs) = Mass of solids (*M*s) /Volume of solids (*V*s)

Particle density of most mineral soils is about 2.6–2.7 g/cm^3. This is because the dominant soil minerals which include quartz, feldspars, micas and clay minerals have an approximate average density of 2.65 g/cm^3.

4.5.1.6 Bulk density

Bulk density measures the physical 'looseness' or 'compaction' of a soil. Dry bulk density is the oven dry weight of soil per unit volume. The dry bulk density expresses the ratio of the mass of solids to the total soil volume while total bulk density is an expression of the total mass of a moist soil per unit volume. Note that bulk density is always smaller than particle density.

Dry bulk density (ρb) = *M*s / *V*t

ρb = *M*s / (*V*s + *V*a + *V*w)

Total bulk density (ρt) = *M*t/*V*t = (*M*s + *M*w) / (*V*s + *V*a + *V*w)

Where *M*s = Mass of solids, *M*w= Mass of water, *M*t = Total mass of soil, *V*t = Total volume of soil, *V*s = Volume of solids, *V*a = Volume of air, *V*w = Volume of water

Total bulk density depends more strongly than does the dry bulk density on soil water content. An ideal soil has a bulk density of about 1.25 g/cm^3. Compacted clay soils have very high bulk densities of between 1.6 and 1.8 g/cm^3.

4.5.1.7 Porosity

Pore space is needed for adequate oxygen supplies to roots, decomposition of organic matter and for storage and infiltration of water. Porosity is an index of the relative pore space in a soil. It is the percentage of a soil that is pore space or voids. Porosity is given by: